

Himanshu Deka Academy

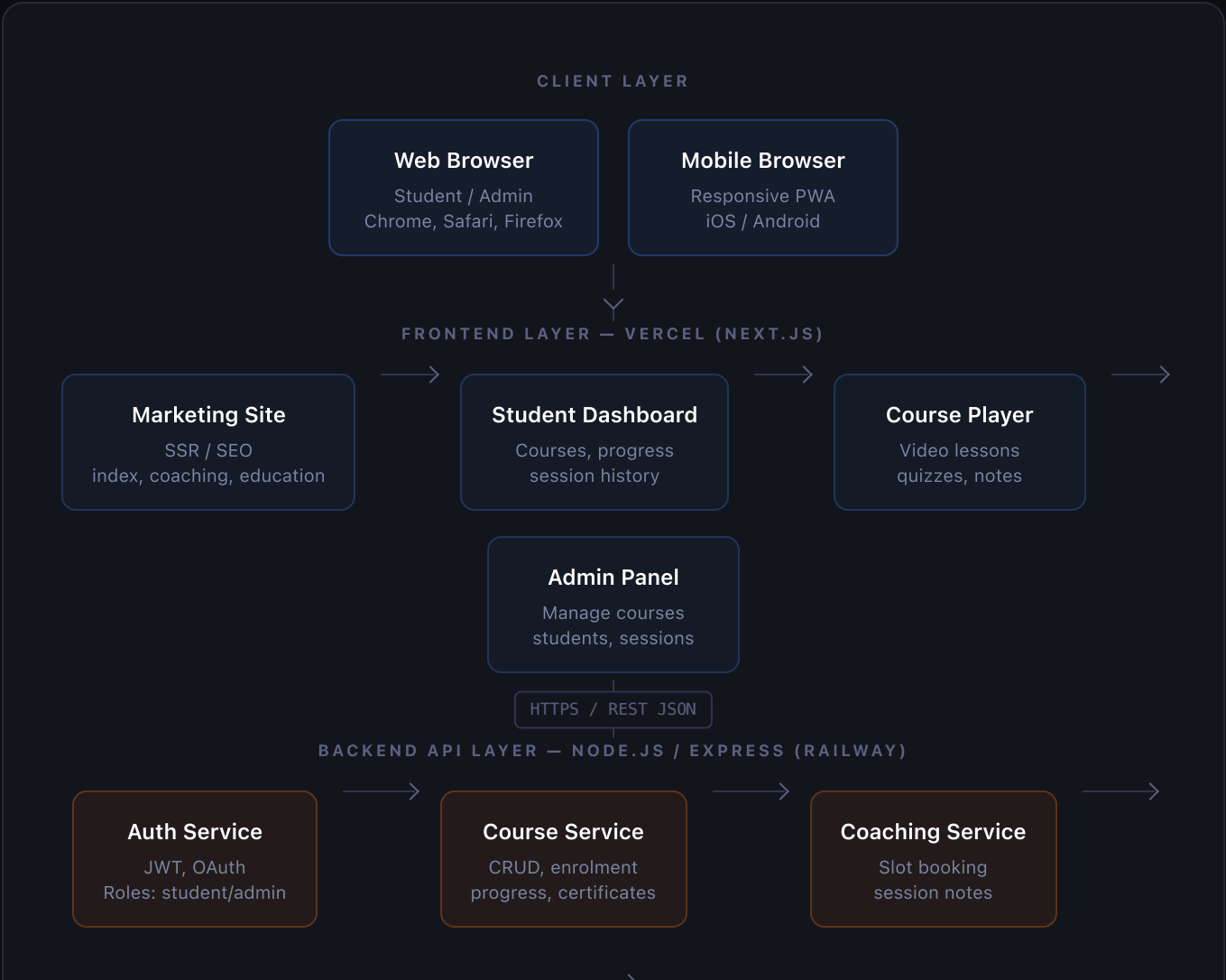
Platform Architecture & Build Roadmap

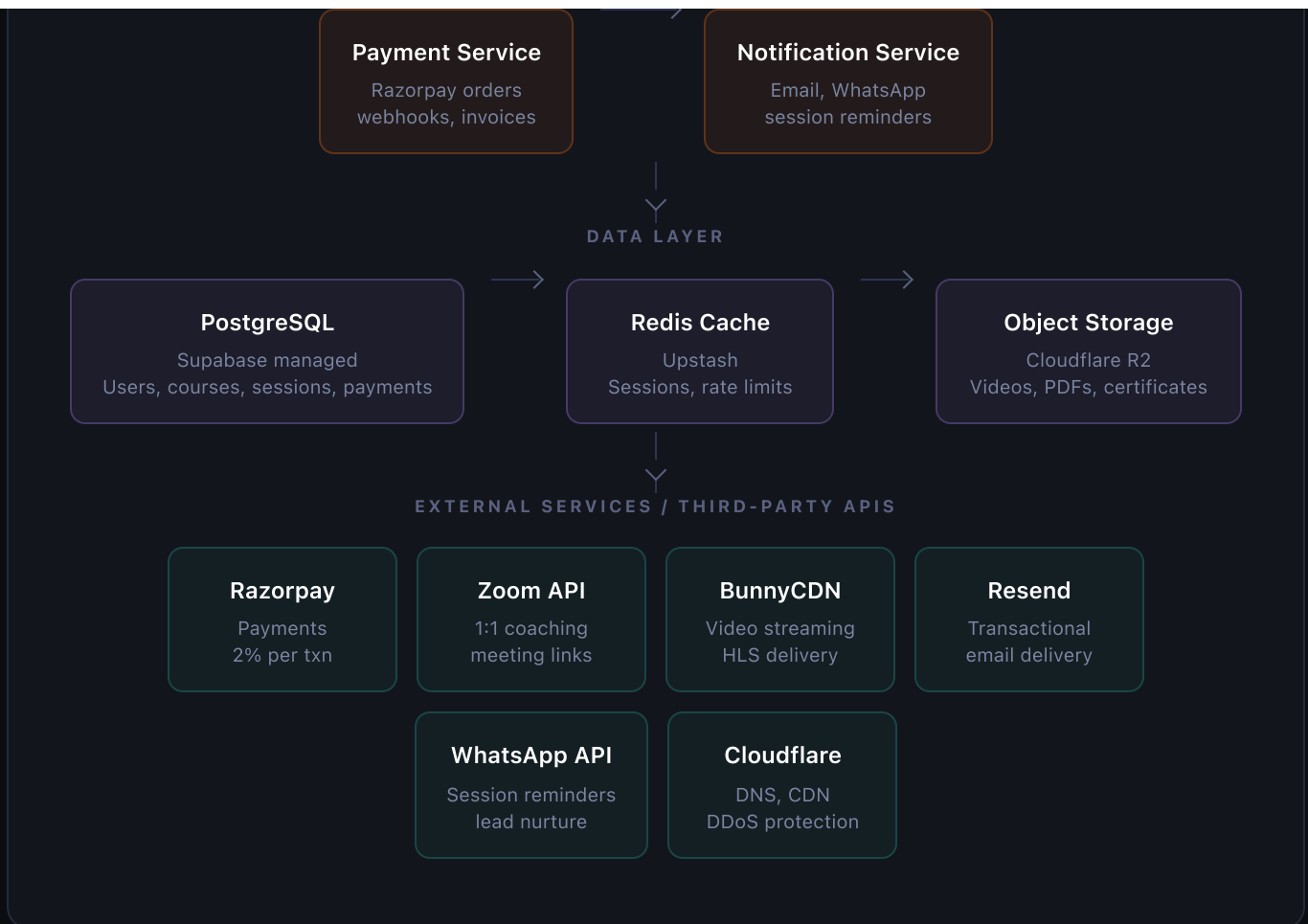
Full-stack architecture for the HDA education platform including course delivery, 1:1 coaching sessions, payment processing, and infrastructure. Estimated total build cost: INR 1.5L – 2.5L.

CURRENT PHASE	TARGET BUILD COST	BUILD TIMELINE	DOCUMENT DATE
• PHASE 1 — STATIC SITE	INR 1,50,000 – 2,50,000	3 – 4 Months	April 2025

01 SYSTEM ARCHITECTURE DIAGRAM

End-to-end data and request flow





02 CURRENT STATE — PHASE 1 (LIVE)

Static website, zero infra cost

Static Website

index.html,
coaching.html,
education.html.
Vanilla HTML / CSS
/ JS. No backend
dependency.
Hosted on GitHub
Pages or Netlify.

Live

GitHub Pages

Netlify

EmailJS

Contact form sends
emails directly from
browser. No server
needed. Free tier
handles 200 emails
per month.

Live

Free

WhatsApp Lead Capture

Apply modal opens
a wa.me deep link
with a pre-filled
message. Zero
backend cost.
Leads managed
manually in a
spreadsheet.

Live

wa.me link

Calendly + Google Form

1:1 booking via embedded
Calendly. Structured intake
via Google Form. Both free.
One pending fix: replace
YOUR_GOOGLE_FORM_LINK
placeholder in js/main.js.

Live

INR 0/mo

03 FRONTEND LAYER — PHASE 2

Next.js 14, Vercel hosting

Marketing Site

Static pages become Next.js for SEO, blog CMS, and testimonials. Server-side rendered for Google indexing.

Phase 2

Next.js

Vercel

INR 0-1,600/mo

Student Dashboard

Post-login view: enrolled courses, progress bars, session history, certificates, upcoming coaching calls.

Phase 2

React

JWT

Course Player

Video lessons with chapter navigation, quizzes, PDF downloads, watch-time tracking, notes. Signed URL access — no public video links.

Phase 2

BunnyCDN

Signed URLs

1:1 Session UI

Book available coaching slots, fill pre-session intake form, view auto-generated Zoom link, access post-session notes.

Phase 2

Zoom SDK

Calendly

04 BACKEND API — PHASE 2

Node.js / Express on Railway

Auth Service

Register, login, Google OAuth, password reset, JWT access + refresh tokens. Role-based: student / admin.

JWT

OAuth2

bcrypt

Course Service

CRUD for courses, modules, lessons. Enrolment. Progress tracking. Quiz engine. PDF certificate generation on completion.

REST

Prisma

Coaching Service

Slot availability, booking creation, session notes, intake form storage, Zoom link auto-generation, rescheduling.

Calendly API

Zoom API

Payment Service

Razorpay order creation, webhook signature verification, refund handling, auto invoice PDF generation via Resend.

Razorpay

Webhooks

Notification Service

Email confirmations, WhatsApp session reminders 24h and 1h before, course progress nudges, certificate delivery.

Resend

WABA

05 DATABASE SCHEMA — POSTGRESQL (SUPABASE)

Core tables · PK = primary key · FK = foreign key

users

Identity

id	uuid	PK
name	varchar(100)	
email	varchar	

courses

Catalogue

id	uuid	PK
title	varchar(200)	
slug	varchar	

modules / lessons

Content hierarchy

module.id	uuid	PK
module.course_id	uuid	
module.title	varchar	

password_hash varchar	description text	lesson.id uuid PK
role enum	price_inr numeric(10,2)	lesson.module_id uuid
phone varchar(20)	duration_weeks int	lesson.video_url varchar
country varchar(5)	level enum	lesson.duration_sec int
created_at timestamp	is_active boolean	lesson.is_preview boolean

enrollments / progress Student tracking	coaching_sessions 1:1 bookings	payments / certificates Transactions + proof
enrollment.id uuid PK	id uuid PK	payment.id uuid PK
enrollment.user_id uuid	student_id uuid FK	payment.user_id uuid
enrollment.course_id uuid	coach_id uuid FK	payment.amount numeric
enrollment.expires_at timestamp	scheduled_at timestamp	payment.razorpay_id varchar
progress.lesson_id uuid	duration_min int	payment.status enum
progress.completed boolean	status enum	cert.enrollment_id uuid
progress.watch_pct int	meet_link varchar	cert.issued_at timestamp
progress.last_watched timestamp	payment_id uuid FK	cert.pdf_url varchar

06

STUDENT LEARNING JOURNEY

End-to-end flow from discovery to certification

01	02	03	04	05	06	07
Discovery Instagram, Google, referral. Lands on marketing site.	Registration Signs up via email or Google OAuth. Intake questionnaire.	Enrolment Selects course or coaching plan. Pays via Razorpay. Invoice emailed.	Self-Study Watches video lessons. Completes quizzes. Tracks progress.	1:1 Coaching Books live session. Joins Zoom. Coach reviews intake and gives plan.	Certification Completes all modules. PDF certificate auto-generated and emailed.	Community Joins WhatsApp alumni group. Referral credit program.

07

1:1 COACHING SESSION FLOW

Booking to follow-up

01	02	03	04	05	06
Student Books Picks available slot from coach calendar	Payment Razorpay processes per-session or bundle fee	Intake Form Pre-session questionnaire: goals, current status	Reminders WhatsApp + email at 24h and 1h before session	Live Session Coach joins Zoom. Reviews form. Real-time guidance.	Follow-up Session notes + action plan emailed. Progress logged.

One-Time Development Cost — Freelancer Route

Single full-stack freelancer (Hyderabad / Bangalore / remote India market) · 3–4 month timeline

PHASE 2 BUILD

COMPONENT	SCOPE	LOW ESTIMATE	HIGH ESTIMATE	NOTES
DESIGN				
UI/UX Design All screens: auth, dashboard, player, admin	Figma, 8–12 screens	INR 12,000	INR 20,000	Can reuse existing site's style. Keep scope tight.
FRONTEND DEVELOPMENT				
Next.js Marketing Site Convert existing HTML pages to Next.js	3 pages + blog	INR 15,000	INR 25,000	Existing design is done. Mostly migration work.
Student Dashboard + Auth UI Login, register, profile, progress view	6–8 screens	INR 18,000	INR 28,000	Zustand for state, React Query for data fetching.
Course Player Video player, chapters, quizzes, notes	1 complex page	INR 12,000	INR 20,000	Use Video.js or Plyr. BunnyCDN HLS stream.
Basic Admin Panel Manage courses, students, bookings	4–6 screens	INR 10,000	INR 18,000	Table views + forms only. No complex analytics.
BACKEND DEVELOPMENT				
Auth Service JWT, Google OAuth, role-based access	API endpoints	INR 10,000	INR 15,000	Use Passport.js. Supabase

Component	Scope	Low Estimate	High Estimate	Notes
				Auth is an option to cut cost.
Course + Enrolment Service CRUD, progress tracking, certificates	Core API	INR 15,000	INR 22,000	Prisma ORM on Supabase PostgreSQL.
Coaching Booking Service Slots, intake forms, Zoom link generation	Booking API	INR 10,000	INR 15,000	Integrate Calendly API or build simple slot system.
Payment Integration Razorpay orders, webhooks, invoices	Payment API	INR 8,000	INR 12,000	Webhook verification is critical. Test thoroughly.
Notification Service Email via Resend, WhatsApp reminders	Background jobs	INR 5,000	INR 8,000	Bull queue on Redis. Simple scheduled jobs.
INFRASTRUCTURE AND DEVOPS				
Database Setup Supabase schema, migrations, seed data	DB config	INR 5,000	INR 8,000	Supabase free tier is sufficient for Phase 2 launch.
Storage + Video CDN Setup Cloudflare R2 bucket, BunnyCDN stream zone	Storage config	INR 5,000	INR 8,000	R2 + BunnyCDN is far cheaper than Mux or Wistia.
Deployment + CI/CD Vercel (frontend), Railway (API), GitHub Actions	DevOps setup	INR 5,000	INR 8,000	One-time. Auto-deploys on every git push after setup.

Component	Scope	Low Estimate	High Estimate	Notes
Domain + SSL hdacoaching.in or himanshudeka.academy	Annual	INR 900	INR 1,500	SSL free via Cloudflare. Domain via GoDaddy / Namecheap.
QUALITY ASSURANCE				
Testing + Bug Fixes Functional testing, payment flow, video playback	QA round	INR 8,000	INR 12,000	Focus on payment and video access — highest risk areas.
Total One-Time Build Cost		INR 1,38,900	INR 2,20,500	Freelancer, 3–4 months. Scope kept lean for Phase 2.

Monthly Infrastructure Cost				RECURRING
Phase 2 (~50–100 students) · All prices in INR per month				
Service	Provider	Monthly Cost	Notes	
Frontend Hosting	Vercel (Hobby / Pro)	INR 0 – 1,600	Free tier covers most traffic at this scale.	
Backend API	Railway Starter	INR 800 – 2,500	Pay-per-use. Scales with requests.	
PostgreSQL Database	Supabase Free / Pro	INR 0 – 2,000	Free tier (500MB) is enough for early stage.	
Redis Cache	Upstash	INR 0 – 800	Pay-per-request. Near zero at low usage.	
Video Storage + CDN	Cloudflare R2 + BunnyCDN	INR 1,500 – 5,000	Depends on content volume and view hours.	
Email	Resend	INR 0 – 1,200	Free 3,000 emails/mo. Pro at INR 1,200/mo unlimited.	

SERVICE	PROVIDER	MONTHLY COST	NOTES
WhatsApp Notifications	Meta WABA via Interakt	INR 500 – 2,000	INR 0.40–0.75 per conversation window (24h).
Zoom (Coaching Sessions)	Zoom Pro — 1 host	INR 1,300	Required for API access to auto-generate meeting links.
Razorpay Gateway Fee	Razorpay	2% per transaction	No monthly fee. On INR 1L revenue = INR 2,000.
Total Monthly Infrastructure		INR 4,100 – 16,400	Excl. Razorpay %. Scales with student count and video usage.

LOW-END TOTAL (YEAR 1)

INR
1,87,900

Build INR 1.39L + Infra INR 4.1k x 12 = INR 49k

MID-RANGE TOTAL (YEAR 1)

INR
2,40,000

Build INR 1.8L + Infra INR 10k x 12 = INR 60k approx

HIGH-END TOTAL (YEAR 1)

INR
4,17,000

Build INR 2.2L + Infra INR 16.4k x 12 = INR 1.97L

09

HOW TO PROCEED — PHASE-BY-PHASE ACTION PLAN

HDA is currently in Phase 1 (static website, building phase)

PHASE 1 — RIGHT NOW (INR 0–5K)

Complete and Launch the Static Site

Validate demand before spending on development. Use no-code tools to collect first revenue.

- 1 Replace `YOUR_GOOGLE_FORM_LINK` in `js/main.js` with your actual Google Form URL
- 2 Buy domain (hdacoaching.in ~INR

PHASE 2 — MONTHS 3–8 (INR 1.5L–2.5L)

Build the Platform After First Revenue

Do not start Phase 2 development until you have earned INR 1L+ from Phase 1 clients. That revenue funds the build.

- 1 Sell first 5–10 coaching packages manually. Use Razorpay link + Zoom + WhatsApp. Confirm people pay.

SHORTCUT BRIDGE (OPTIONAL)

No-Code LMS While Platform is Being Built

If you cannot wait 3–4 months for a custom build, use Teachable as a bridge. Go live in one week.

- A **Teachable** — Full LMS with video, quizzes, certificates, Razorpay. **INR 3,500–7,000/mo.** Launch in 1 week. No dev needed.

900) and deploy to Netlify or GitHub Pages

- 3 Set up a **Razorpay Payment Link** for each coaching package. No backend needed — share via WhatsApp
- 4 Connect your real **Calendly** link into the modal. Set your availability for 1:1 sessions
- 5 Add **Google Analytics 4** to the site. You need traffic data before deciding where to invest

- 2 Record 3–5 course modules. Sell via Razorpay link + Google Drive delivery. Validate course demand.
- 3 Hire a **full-stack freelancer** (Upwork / LinkedIn). Budget INR 1.4L–2.2L for the full build per the cost table above.
- 4 Set up Supabase (DB), Railway (API), Vercel (frontend), R2 + BunnyCDN (video). Use free tiers first.
- 5 Launch with 1 course + 1:1 coaching. Do not wait for a perfect product — ship and iterate.

B Kajabi — Courses + email marketing + community + website builder. **INR 11,000–25,000/mo**. Best all-in-one for a coaching business.

C Custom Platform — Full control, lower long-term cost, own branding. **INR 1.5L–2.5L one-time**. 3–4 months build.

Recommendation: Use **Teachable as a bridge** to start earning while the custom platform is being built. Migrate students once the custom build is live.

Immediate Action Checklist — Do This Week

01 Replace **YOUR_GOOGLE_FORM_LINK** in `js/main.js` with your real Google Form URL

02 Buy **hdacoaching.in** on GoDaddy (~INR 900/yr). Connect to Netlify. Enable Cloudflare for free SSL and CDN.

03 Create a **Razorpay Payment Link** for your 1:1 coaching package. Add it to the site CTA and your WhatsApp bio.

04 Set up **Calendly** with your real availability. Replace the placeholder link in the apply modal.

05 Add **Google Analytics 4** to the site. 5 minutes. You need traffic source data before making any marketing decisions.

06 Decide: **Teachable bridge** (live in 1 week, INR 3.5k/mo) or wait for custom build (INR 1.5L–2.5L, 3–4 months).

10 REVENUE MODEL

Illustrative targets at scale

INR 5k–12k

INR 15k–60k

INR 2k–5k

10–20%
Affiliate
Commission

**Per Course
Enrolment**

Self-paced video course. 100 students/quarter = INR 5L–12L revenue

**Per Coaching
Package**

4–24 week 1:1 mentorship. 10 clients/month = INR 1.5L–6L/mo

**Monthly
Subscription**

All-access pass. 50 members = INR 1L–2.5L MRR

Student referral credits. Alumni network creates a marketing flywheel.

11

FULL TECH STACK

Chosen for cost efficiency at this scale

FRONTEND

Next.js 14
React
Tailwind CSS
Zustand
React Query

BACKEND

Node.js
Express
Prisma ORM
Bull Queue
Zod

DATA

PostgreSQL
Supabase
Upstash Redis
Cloudflare R2

DEVOPS

GitHub Actions
Vercel
Railway
Cloudflare
Sentry

• **PHASE 1 — NOW**

Static Website

HTML/CSS/JS. Lead capture via WhatsApp + EmailJS + Calendly. Razorpay Payment Links for manual payments.

index.html
coaching.html
INR 0/mo infra

PHASE 2 — Q3–Q4 2025

Platform Launch

Auth, course player, 1:1 booking system, Razorpay checkout, student dashboard. First 3–5 courses live.

Backend API
Course Player
INR 4k–16k/mo

PHASE 3 — 2026+

Scale and Community

Live cohort batches, alumni community, affiliate program, mobile app, corporate B2B packages.

Mobile App
Cohorts
INR 28k–77k/mo

Notes: All INR figures are estimates as of April 2025. Development cost assumes a single experienced full-stack freelancer from the Indian market (Hyderabad, Bangalore, or remote). Agency rates are 2x–4x higher. Razorpay transaction fee (2%) is variable and excluded from monthly infra totals. Video streaming costs depend heavily on content volume and student viewing hours — budget INR 5–15 per GB of egress. For Phase 2, starting with Supabase (database), Railway (API), Vercel (frontend), Cloudflare R2 (storage), and Resend (email) gives the best cost-to-capability ratio for a bootstrapped launch. Migrate to dedicated infrastructure only when monthly revenue exceeds INR 3L consistently.